Sürekli zamanlı Fourier Serisinin Özellikleri

$x(t) = x(t+T_0)$, $\omega_0 = 2\pi f_0 = \frac{2\pi}{T_0}$

$x(t) \xleftrightarrow{FS} x_k$

1) **Scalar multiplication:** $\alpha x(t) \xleftrightarrow{FS} \alpha x_k$

2) **Linearlik:** $\alpha x_1(t) + \beta x_2(t) \xleftrightarrow{FS} \alpha x_{1,k} + \beta x_{2,k}$

$x_1(t)$ ve $x_2(t)$'nin periyodu (T_0) / frekansları (ω_0) aynı olmalıdır.

$\alpha x_1(t) + \beta x_2(t)$ işaretinin periyodu da T_0'dır.

3) **Zamanda tersini alma (time reversal):** $x(-t) \xleftrightarrow{FS} x_{-k}$

$\Rightarrow x(t)$ çift işaret ise FS katsayıları da çifttir.

$x(t) = x(-t) \xleftrightarrow{FS} x_k = x_{-k}$

$\Rightarrow x(t)$ tek işaret ise FS katsayıları da tektir.

$x(t) = -x(-t) \xleftrightarrow{FS} x_k = -x_{-k}$

4) **Zamanda öteleme:** $x(t-t_0) \xleftrightarrow{FS} x_k e^{-jk\omega_0 t_0}$

İspat: $x(t-t_0) \xleftrightarrow{FS} d_k$ olsun

$$d_k = \frac{1}{T_0}\int_{T_0} x(t-t_0)\, e^{-jk\omega_0 t}\, dt \underset{t-t_0=\tau}{=} \frac{1}{T_0}\int_{T_0} x(\tau)\, e^{-jk\omega_0(\tau+t_0)}\, d\tau = e^{-jk\omega_0 t_0}\underbrace{\frac{1}{T_0}\int_{T_0} x(\tau)\, e^{-jk\omega_0\tau}\, d\tau}_{=x_k}$$

$$d_k = x_k e^{-jk\omega_0 t_0}$$

$|d_k| = |x_k|$ → Periyodik bir işaretin zamanda ötelenmesi FS katsayılarının genliğini değiştirmez.

5) **Frekansta öteleme:** $x(t)e^{jk_0\omega_0 t} \xleftrightarrow{FS} x_{k-k_0}$

İspat: $d_k = \frac{1}{T_0}\int_{T_0} x(t) e^{jk_0\omega_0 t} \cdot e^{-jk\omega_0 t}\, dt = \frac{1}{T_0}\int_{T_0} x(t) e^{-j(k-k_0)\omega_0 t}\, dt = x_{k-k_0}$

6) **Zamanda ölçeklendirme:** $x(\alpha t) \xleftrightarrow{FS} x_k$

Periyodik bir işaretin zamanda ölçeklendirilmesi FS katsayılarını değiştirmez. Ancak, $x(\alpha t)$ işaretinin frekansı $\alpha\omega_0$ ve periyodu $\frac{T_0}{\alpha}$ olarak değişir.

7) **Çarpma:** $x_1(t).x_2(t) \xleftrightarrow{FS} x_{1,k} * x_{2,k} = \sum_{n=-\infty}^{\infty} x_{1,n}\, x_{2,k-n}$
(Frekansta konvolüsyon)

8) **Zamanda türev alma:** $\frac{d^n x(t)}{dt^n} \xleftrightarrow{FS} (jk\omega_0)^n x_k$

9) **Eşlenik alma (conjugation):** $x^*(t) \xleftrightarrow{FS} x^*_{-k}$

İspat: $x^*(t) = \sum_{k=-\infty}^{\infty} x_k^* e^{-jk\omega_0 t} \underset{k=-n}{=} \sum_{n=-\infty}^{\infty} x^*_{-n} e^{jn\omega_0 t}$

⇒ $x(t)$ işareti reel ise FS katsayıları eşlenik simetriktir.

$$x(t) = x^*(t) \xleftrightarrow{FS} x_k = x_{-k}^* \longrightarrow x_0 \text{ reeldir.}$$

$$|x_k| = |x_{-k}|$$

$$\theta_k = -\theta_{-k} \quad (\angle x_k = -\angle x_{-k})$$

⇒ $x(t)$ işareti reel ve çift ise FS katsayıları da reel ve çifttir.

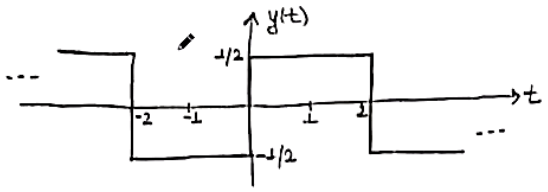
$$x(t) = x^*(t) = x(-t) \xleftrightarrow{FS} x_k = x_{-k}^* = x_{-k} \longrightarrow x_k^* = x_k$$

⇒ $x(t)$ işareti reel ve tek ise FS katsayıları sanal ve tektir.

$$x(t) = x^*(t) = -x(-t) \xleftrightarrow{FS} x_k = x_{-k}^* = -x_{-k} \longrightarrow x_k^* = -x_k$$

$$\hookrightarrow x_0 = 0$$

Ex:

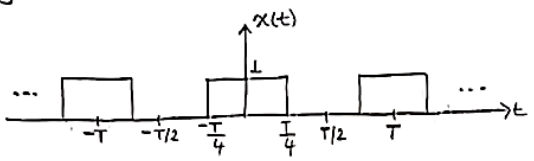


$y(t)$ işaretinin Fourier Serisi katsayılarını bulunuz.

I. yol: $y(t) = \sum_{k=-\infty}^{\infty} y_k e^{jk\omega_0 t}$, $T_0 = 4$, $\omega_0 = \frac{2\pi}{T_0} = \frac{\pi}{2}$

$$y_k = \frac{1}{T_0} \int_{T_0} y(t) e^{-jk\omega_0 t} dt$$

II. yol:



$$x_0 = \frac{1}{2}$$

$$x_k = \frac{\sin(k\pi/2)}{\pi k}, \quad k \neq 0$$

$T = 4 \Rightarrow y(t) = x(t-1) - 1/2$

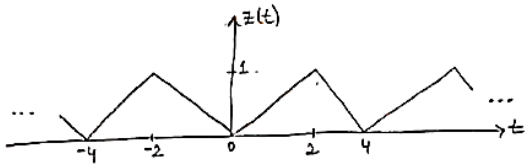
$$y_k = \begin{cases} x_0 - 1/2, & k = 0 \\ x_k e^{-jk\omega_0 t_0} = x_k e^{-jk\pi/2}, & k \neq 0 \end{cases}$$

← $-1/2$ dc offset

← $x(t-1)$ zamanda kaydırma.

$$y_k = \begin{cases} 0, & k = 0 \\ \dfrac{\sin(k\pi/2)\, e^{-jk\pi/2}}{k\pi}, & k \neq 0 \end{cases}$$

Ex:



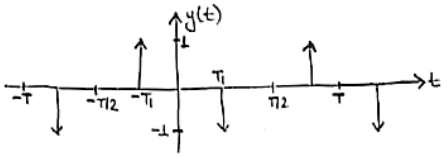
$z(t) = |t|/2, \quad |t|<2$

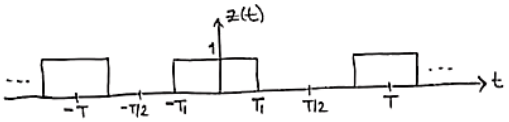
$\frac{dz(t)}{dt} = y(t) \xleftrightarrow{FS} y_k = jk\omega_0 z_k \Rightarrow z_k = \frac{1}{jk\omega_0} y_k = \frac{2}{jk\pi} y_k$ ($\omega_0 = \pi/2$)

$z_k = \frac{2}{j(k\pi)^2} \sin(k\pi/2) e^{-jk\pi/2}, \quad k \neq 0$

$z_0 = \frac{1}{T_0} \int_{T_0} z(t)\,dt = \frac{1}{2}$

Ex: $x(t) = \sum_{k=-\infty}^{\infty} \delta(t-kT)$ dürtü treni olup $x_k = \frac{1}{T}$'dir. Aşağıda verilen işaretlerin FS katsayılarını bulunuz.





$y(t) = x(t+T_1) - x(t-T_1) \xleftrightarrow{FS} y_k = x_k e^{jk\omega_0 T_1} - x_k e^{-jk\omega_0 T_1} = \frac{1}{T}\left[e^{+jk\omega_0 T_1} - e^{-jk\omega_0 T_1}\right]$

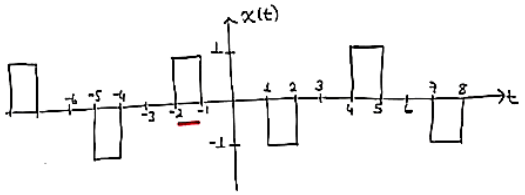
$$y_k = \frac{2j}{T}\sin(k\omega_0 T_1)$$

$\frac{dz(t)}{dt} = y(t) \xleftrightarrow{FS} y_k = jk\omega_0 z_k \longrightarrow z_k = \frac{1}{jk\omega_0} y_k = \frac{2j}{Tjk\omega_0}\sin(k\omega_0 T_1)$

$$z_k = \frac{\sin(k\omega_0 T_1)}{k\pi}, \quad k \neq 0$$

$z_0 = \frac{1}{T_0}\int_{T_0} z(t)\,dt = \frac{1}{T}\int_{-T_1}^{T_1} dt = \frac{2T_1}{T}$

Ex:



$x(t) = z(t+3/2) - z(t-3/2)$ ve $T_1 = 1/2$

$x_k = z_k e^{+jk\omega_0 3/2} - z_k e^{-jk\omega_0 3/2}$, $T=6 \longrightarrow \omega_0 = \frac{2\pi}{T} = \frac{\pi}{3}$

$x_k = z_k\left[e^{jk\pi/2} - e^{-jk\pi/2}\right] = 2j z_k \sin(k\pi/2)$

$x_k = 2j \frac{1}{k\pi}\sin(k\pi/6)\sin(k\pi/2)$

$$x_k = \frac{\cos(2k\pi/3) - \cos(k\pi/3)}{jk\pi}$$

Sürekli Zamanlı LTI sistemlerin Periyodik İşaretlere Cevabı

$$Ae^{st} \longrightarrow \boxed{h(t)} \longrightarrow Ae^{st}\cdot H(s), \quad H(s)=\int_{-\infty}^{\infty} h(\tau)e^{-s\tau}d\tau$$

⇒ Bir sistemin çıkışı, girdinin bir sabit ile çarpımı ise, o girdi işaretine sistemin özfonksiyonu (eigenfunction), ilgili sabite (genlik çarpanına) özdeğeri (eigenvalue) denir.

$$y(t)=x(t)*h(t)=\int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau=\int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau$$

$$y(t)=\int_{-\infty}^{\infty} h(\tau)Ae^{s(t-\tau)}d\tau=Ae^{st}\underbrace{\int_{-\infty}^{\infty} h(\tau)e^{-s\tau}d\tau}_{\triangleq H(s)}=Ae^{st}H(s) \longrightarrow Ae^{st}: \text{öz-fonksiyon}$$

$\hookrightarrow H(s)$: özdeğer

⇒ $s=j\omega$

$$Ae^{j\omega t} \longrightarrow \boxed{h(t)} \longrightarrow Ae^{j\omega t}H(j\omega)$$

$$H(j\omega)=\int_{-\infty}^{\infty} h(t)e^{-j\omega t}dt : \text{Frequency Response (Frekans cevabı)}$$

$$x(t)=\sum_{k=-\infty}^{\infty} x_k e^{jk\omega_0 t} \longrightarrow \boxed{h(t)} \longrightarrow y(t)=\sum_{k=-\infty}^{\infty} \underbrace{x_k H(jk\omega_0)}_{\triangleq y_k} e^{jk\omega_0 t}=\sum_{k=-\infty}^{\infty} y_k e^{jk\omega_0 t}$$

Bir LTI sistemin girişinde hangi frekans bileşenleri varsa çıkışında da o frekans bileşenleri vardır.

Ex: $x(t)=\sum_{k=-3}^{3} x_k e^{jk2\pi t}$, $x_0=1$, $x_1=x_{-1}=\frac{1}{4}$, $x_2=x_{-2}=\frac{1}{2}$, $x_3=x_{-3}=\frac{1}{3}$

a) $x(t)=?$

b) $h(t)=e^{-t}u(t)$ ise $y(t)=?$

a) $$x(t)=1+\frac{1}{4}\left[e^{j2\pi t}+e^{-j2\pi t}\right]+\frac{1}{2}\left[e^{j4\pi t}+e^{-j4\pi t}\right]+\frac{1}{3}\left[e^{j6\pi t}+e^{-j6\pi t}\right]$$

$$x(t)=1+\frac{1}{2}\cos(2\pi t)+\cos(4\pi t)+\frac{2}{3}\cos(6\pi t)$$

b) $$y(t)=\sum_{k=-\infty}^{\infty} y_k e^{jk\omega_0 t}=\sum_{k=-\infty}^{\infty} x_k H(jk\omega_0)e^{jk\omega_0 t}, \quad \omega_0=2\pi$$

$$H(j\omega)=\int_{-\infty}^{\infty} h(t)e^{-j\omega t}dt=\int_{0}^{\infty} e^{-t}\cdot e^{-j\omega t}dt=\int_{0}^{\infty} e^{-(1+j\omega)t}dt=\frac{e^{-(1+j\omega)t}}{-(1+j\omega)}\Big|_0^{\infty}=\frac{1}{1+j\omega}$$

$$y_k=x_k H(jk\omega_0)=x_k H(jk2\pi)$$

$$y_0=x_0=1$$

$$y_1=x_1 H(j2\pi)=\frac{1}{4}\left(\frac{1}{1+j2\pi}\right)$$

$$y_{-1}=x_{-1}H(-j2\pi)=\frac{1}{4}\left(\frac{1}{1-j2\pi}\right)$$

$$y_2=\frac{1}{2}\left(\frac{1}{1+j4\pi}\right) \qquad y_{-2}=\frac{1}{2}\left(\frac{1}{1-j4\pi}\right) \qquad y_3=\frac{1}{3}\left(\frac{1}{1+j6\pi}\right) \qquad y_{-3}=\frac{1}{3}\left(\frac{1}{1-j6\pi}\right)$$

Ex: $x(t)=\cos(2t+\pi/4)$, $x_k=?$ $P_x=?$

$x(t)=\frac{1}{2}\left[e^{j(2t+\pi/4)}+e^{-j(2t+\pi/4)}\right]=\frac{1}{2}e^{j\pi/4}\cdot e^{j2t}+\frac{1}{2}e^{-j\pi/4}\cdot e^{-j2t}$, $\omega_0=2 \rightarrow T=\frac{2\pi}{\omega_0}=\pi$

$x_1=\frac{1}{2}e^{j\pi/4}$ $\quad x_{-1}=\frac{1}{2}e^{-j\pi/4}$

$P_x=\frac{1}{T}\int_T |x(t)|^2 dt=\sum_{k=-\infty}^{\infty}|x_k|^2=\left(\frac{1}{2}\right)^2\cdot 2=\frac{1}{2}$

Ex: $x_k=\begin{cases} jk, & |k|\leq 3 \\ 0, & \text{diğerinde}\end{cases}$ ise $x(t)=?$

$x(t)=\sum_{k=-3}^{3} x_k e^{jk\omega_0 t} \longleftarrow x_0=0$

$x_1=j \quad x_{-1}=-j$

$x_2=2j \quad x_{-2}=-2j$

$x_3=3j \quad x_{-3}=-3j$

$x(t)=j\left[e^{j\omega_0 t}-e^{-j\omega_0 t}\right]+2j\left[e^{j2\omega_0 t}-e^{-j2\omega_0 t}\right]+3j\left[e^{j3\omega_0 t}-e^{-j3\omega_0 t}\right]$

$x(t)=-2\sin\omega_0 t-4\sin(2\omega_0 t)-8\sin(3\omega_0 t)$

Ex: $H(j\omega)=\begin{cases}1, & |\omega|\leq 100 \\ 0, & |\omega|>100\end{cases}$ $\quad x(t)$'nin periyodu $T=\pi/6$

$y(t)=x(t)$ olması için hangi k değerleri için $x_k=0$ olmalıdır.

$y(t)=\sum_{k=-\infty}^{\infty} y_k e^{jk\omega_0 t}=\sum_{k=-\infty}^{\infty} x_k H(jk\omega_0)e^{jk\omega_0 t}$

$|k\omega_0|\leq 100$

$\omega_0=\frac{2\pi}{T}=\frac{2\pi}{\pi/6}=12$

$\Big\} \; |k|\leq \frac{100}{12}=8.3 \rightarrow |k|\leq 8 \rightarrow |k|>8$

★ Ex: Bir $x(t)$ işareti, aşağıda listelenen özellikleri ile (T işareti dışında) belirlenebilir mi?

1. Reeldir.
2. $T_0=4$ periyodu ile periyodiktir ve FS katsayıları x_k'dır.
3. $x_k=0$, $|k|>1$.
4. FS katsayıları $y_k=e^{-jk\pi/2}x_{-k}$ olan $y(t)$ sinyali tektir.
5. $\frac{1}{4}\int_4 |x(t)|^2 dt=1/2$.

2. $T_0=4 \rightarrow \omega_0=\frac{2\pi}{T_0}=\frac{\pi}{2}$

3. $x_k=0$, $|k|>1 \longrightarrow x(t)=\sum_{k=-1}^{1} x_k e^{jk\omega_0 t}=x_0+x_1 e^{jk\pi t/2}+x_{-1}e^{-jk\pi t/2}$

1. $x(t)$ Reel $\longrightarrow x(t)=x^*(t) \Rightarrow x_k=x_{-k}^*$ ve x_0 Reel

$x(t)=x_0+x_1 e^{jk\omega_0 t}+\left(x_1 e^{jk\omega_0 t}\right)^*=x_0+2\,\mathrm{Re}\left\{x_1 e^{jk\pi t/2}\right\}$

4. $y_k=e^{-jk\pi/2}\cdot x_{-k} \longrightarrow x(t)\overset{FS}{\longleftrightarrow} x_k$ ise $x(-t)\overset{FS}{\longleftrightarrow} x_{-k}$

$x(t-t_0)\overset{FS}{\longleftrightarrow} x_k e^{-jk\omega_0 t_0}$

$y(t)=x(-t+1)$ reel ve tektir: $\longrightarrow y_k$ sanal ve tek $\longrightarrow y_0=0$

$\hookrightarrow y_k=-y_{-k}$ $\quad (y_1=-y_{-1})$

5. $\frac{1}{4}\int_4 |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |x_k|^2 = \sum_k |y_k|^2 = |y_0|^2 + |y_1|^2 + |y_{-1}|^2$

$2|y_1|^2 = \frac{1}{2} \rightarrow |y_1| = \frac{1}{2}$

y_k sonal olacağından $y_k = \mp j/2$

$x_{-k} = y_k e^{jk\pi/2}$

- $\rightarrow x_0 = 0$
- $\rightarrow y_1 = j/2 \Rightarrow x_{-1} = \frac{j}{2} e^{j\pi/2} = -\frac{1}{2}$, $x_{-1}^* = x_1 = \frac{-1}{2}$ $\Big\}$ $x(t) = 2\,Re\left\{\frac{-1}{2} e^{j\pi t/2}\right\} = -\cos(\pi t/2)$
- $\rightarrow y_{-1} = -j/2 \Rightarrow x_{-1} = \frac{1}{2} = x_1 \longrightarrow x(t) = \cos(\pi t/2)$